

Predicting Diabetes Risk

The dataset:

The dataset for Diabetes Risk was downloaded website Kaggle and originally consisted of 15,000 patients but we reduced to about 6,500 patients from six Indian cities to be within the scope of the class project. Each row is one patient described by 18 features.

Source URL: <https://www.kaggle.com/datasets/mansiaggarwal88/diabetes-risk-prediction>

Why we chose it:

- This is a clear example of a supervised classification problem because each record has a labeled target (diabetes_risk column).
- It is a good example of real-world importance and application. According to the International Diabetes Federation by 2050 their projections show that 1 in 8 adults or approximately 853 million will be living with diabetes. A model that uses this kind of data could help screening and risk.
- It is a good size dataset to practice the pre-processing steps we have learned in class.

What we will accomplish:

- Split data: Use Python, to divide the data 70% for training and 30% for testing. Perform EDA only on the training set.
- NaN and Null Values: Find the features that missing values and fill them in with the mean, median, or mode. Looking at the data briefly Alcohol consumption column is probably one that will need this.
- Encoding: Since models only understand numbers, columns with order change to numbers. Ex., Physical Activity Level, Family History Diabetes, Diabetes Risk.
- One-hot Encoding: Columns with no order get 0/1 column for each category so the model does not think one is “higher” than the other. Ex., City, Gender, Diet-type.
- Scaling: Since blood pressure and stress have a wide range Min-Max scaling will be used to put the features in a 0-1 scale and keep the numbers in closer range.